Sizhe Li

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EDUCATION

The Chinese University of Hong Kong, Shenzhen *Shenzhen, China*
Bachelor of Science in Statistics *Sep. 2022 – Present*
Cumulative GPA: 3.92/4.00, Major GPA: 3.98/4.00

University of California, Berkeley *California, USA*
Visiting Student *Aug. 2024 – Dec. 2024*
GPA: 4.00/4.00

No.1 Middle School Affiliated to Central China Normal University *Wuhan, China*
High School Diploma *Sep. 2019 – Jun. 2022*

RESEARCH EXPERIENCE

Prediction-Specific Design of Learning-Augmented Algorithms *Feb 2024 – May 2025*
Supervised by Prof. Tongxin Li and Prof. Nicolas Christianson

- **New Framework & Definitions:** Introduced a *prediction-specific framework* for learning-augmented algorithms enabling fine-grained, per-prediction performance guarantees. Proposed a novel definition of *strong optimality* capturing Pareto optimality in both consistency and robustness tradeoffs. Proved the suboptimality of state-of-the-art methods (NeurIPS 2018, 2021) in this new framework.
- **Generic Approach:** Developed a *bi-level optimization* framework for designing strongly-optimal algorithms, applicable to diverse online problems.
- **Explicit Algorithm Design:** Designed provably strongly optimal algorithms for *deterministic ski rental*, *randomized ski rental* and *one-max search*.

Optimal Tradeoff in Learning-Augmented Non-Clairvoyant Scheduling *June 2025 – Present*
Supervised by Prof. Tongxin Li and Prof. Nicolas Christianson

- Ongoing work on the consistency–robustness trade-off in n-job non-clairvoyant scheduling. Alex Wei & Fred Zhang (2020) present a lower bound that is only proven tight for $n = 2$ and at two endpoints, and the optimal trade-off for general n remains open. This work shows the Wei's bound is not tight when $n \geq 5$ and gives a refined (near-)optimal lower bound.

PUBLICATIONS

- **Sizhe Li**, Nicolas Christianson, Tongxin Li. *Prediction-Specific Design of Learning-Augmented Online Algorithms.* arXiv:2510.14887, arXiv preprint, under review at *ACM SIGMETRICS.*

PROJECTS

Simulation Model for ICU Resource Optimization, UC Berkeley *Aug. 2024 – Dec. 2024*

- Developed a discrete-event simulation to optimize ICU capacity using the MIMIC-IV demo dataset.
- Proposed patient prioritization based on severity and urgency, using dynamic penalty evaluation.

- Designed and compared three resource allocation solvers under varied load conditions.
- Conducted sensitivity analysis to evaluate model robustness and key performance drivers.

TEACHING

Undergraduate Teaching Fellow, CUHK-Shenzhen

- **MAT2041: Linear Algebra and Applications** Fall 2023, Spring 2025
- **CSC1001: Introduction to Computer Science: Programming Methodology** Fall 2025

HONORS AND AWARDS

- Dean's List, 2023–2025
- Academic Performance Scholarship (Top 1–3%), 2023–2024
- Undergraduate Research Award, 2025

SKILLS

- **Programming:** Python (NumPy, Pandas, SciPy), Java, R, MATLAB
- **Tools:** Git, Jupyter, Linux, Overleaf
- **Typesetting:** LaTeX